

Supplementary Table S1: Average minimum log differences required for a significant result with ALDEx2 and ALDEx3 for all datasets and γ values. For all datasets, the mean minimum log difference required for ALDEx2 and ALDEx3 to recognise a feature as differentially expressed (i.e. BH-adjusted P -value < 0.05) was calculated for γ values ranging from 0 to 0.5. Data represent a mean of all 100 iterations of permuted/thinned HTS count data (TCGA, The Cancer Genome Atlas).

Dataset	Scale (γ)	Mean minimum difference ($\pm$ s.d.)	
		ALDEx2	ALDEx3
Immunotherapy	0	0.34 (0.045)	0.36 (0.041)
	0.1	0.54 (0.044)	0.48 (0.040)
	0.2	0.78 (0.059)	0.64 (0.040)
	0.3	1.02 (0.079)	0.80 (0.051)
	0.4	1.27 (0.108)	0.98 (0.061)
	0.5	1.52 (0.132)	1.14 (0.070)
Metatranscriptome	0	0.96 (0.261)	0.96 (0.260)
	0.1	1.10 (0.246)	1.04 (0.257)
	0.2	1.35 (0.280)	1.21 (0.245)
	0.3	1.59 (0.275)	1.37 (0.271)
	0.4	1.90 (0.327)	1.57 (0.291)
	0.5	2.16 (0.376)	1.76 (0.291)
Single-cell	0	0.85 (0.213)	0.93 (0.250)
	0.1	1.00 (0.202)	1.01 (0.233)
	0.2	1.21 (0.191)	1.18 (0.230)
	0.3	1.45 (0.199)	1.34 (0.222)
	0.4	1.73 (0.208)	1.53 (0.225)
	0.5	2.01 (0.253)	1.72 (0.235)
TCGA	0	0.24 (0.056)	0.25 (0.055)
	0.1	0.45 (0.051)	0.39 (0.049)
	0.2	0.69 (0.068)	0.55 (0.053)
	0.3	0.93 (0.089)	0.72 (0.060)
	0.4	1.17 (0.113)	0.88 (0.068)
	0.5	1.41 (0.140)	1.05 (0.077)
Tianyi 16S rRNA	0	1.81 (0.551)	1.91 (0.552)
	0.1	1.89 (0.523)	1.96 (0.525)
	0.2	2.07 (0.497)	2.07 (0.549)
	0.3	2.30 (0.518)	2.19 (0.533)
	0.4	2.50 (0.507)	2.35 (0.519)
	0.5	2.79 (0.684)	2.61 (0.632)
Yeast	0	0.18 (0.046)	0.18 (0.046)
	0.1	0.39 (0.045)	0.32 (0.040)
	0.2	0.63 (0.065)	0.47 (0.044)
	0.3	0.87 (0.087)	0.65 (0.056)
	0.4	1.11 (0.113)	0.81 (0.062)
	0.5	1.35 (0.147)	0.98 (0.069)